Print-Ready Documents from HTML

How teams turn web content into reports, invoices and catalogs that look as good on paper as they do on screen.

1. Why HTML as a source format

Most business documents already exist as HTML somewhere: an order confirmation in a web shop, a report in a dashboard, a contract in a portal. Generating the PDF from the same markup avoids a second, separately maintained template in a word processor or a reporting tool, and keeps the printed version in step with the screen version.

HTML and CSS also bring a large pool of skills with them. Designers who build web pages can style a document without learning a proprietary layout language, and developers can fill the template with data using the tools they already use for the web front end.

The price for this convenience is that the web was not made for paper. A browser window has no pages, no margins and no running headers. A PDF converter has to add these concepts, and it has to decide where a page ends without cutting a line of text or a table row in half.

2. Pagination

Page breaks are the most visible part of the difference between screen and paper. CSS offers the *page-break-before*, *page-break-after* and *page-break-inside* properties, and a converter may add its own rules on top of them: keeping a heading together with the paragraph that follows it, repeating the header row of a long table on every page, or avoiding a single orphaned line at the bottom of a page.

Headers and footers are the second part. They carry the page number, the document title or a confidentiality note, and they often change between the first page and the following ones. In PD4ML they are defined either through the API or inline, with the *<pd4ml:page.header>* and *<pd4ml:page.footer>* tags.

3. Fonts and text

A PDF is only portable if the fonts travel with it. Embedding the TrueType fonts used by the document guarantees that a reader on another computer sees the same glyphs, line breaks and page count. Subsetting keeps the file small: only the characters actually used are embedded.

Fonts are also where internationalization starts. Cyrillic, Greek, Arabic, Hebrew or CJK text needs a font that covers these scripts, and right-to-left languages need the text direction handled correctly in every paragraph and table cell.

Script	Example font	Direction
Latin, Cyrillic, Greek	Noto Sans	left to right
Arabic	Noto Naskh Arabic	right to left
Hebrew	Noto Sans Hebrew	right to left
Chinese, Japanese, Korean	Noto Sans CJK	left to right

4. Standards for archiving and invoicing

Documents that have to be readable in twenty years are usually stored as PDF/A, an ISO standard that forbids everything a future reader might not be able to resolve: external fonts, encryption, scripts. PDF/A-3 additionally allows embedded files, which is the foundation of the ZUGFeRD and Factur-X e-invoicing formats.[1]

Accessibility is the other standard to watch. PDF/UA requires a tagged document structure, so that screen readers can follow headings, lists and tables in the right order. Converting from well-structured HTML is a good starting point: the semantic markup already describes most of what the tags need to express.

5. Working with existing PDFs

Not every task starts with HTML. Merging a generated cover letter with a contract that already exists as PDF, adding a digital signature, filling in or exporting form data, or reducing the file size before archiving are all operations on existing documents.

Signatures deserve a special mention: a signed PDF proves who approved the document and that it has not been changed since. Long-term validation information keeps the signature verifiable after the signing certificate has expired.

6. Layout in columns

Newsletters, catalogs, glossaries and price lists are easier to read in narrow columns than in lines that span the full page width. This newsletter is an example: PD4ML lays the document out to the width of a column, fills the columns of each page one after another and starts a new page after the last column.

Headers, footers, backgrounds and watermarks still belong to the page, so the header above spans both columns, and the page numbers count pages, not columns. A footnote, on the other hand, is placed at the bottom of the column that refers to it.

[1] ZUGFeRD and Factur-X embed the invoice as XML in a PDF/A-3 document, so the same file serves both people and accounting software.

Appendix: How this newsletter was made

The document is an ordinary HTML page with a page header, a page footer and a footnote. The Java code converts it with two columns per page and an 8 millimeter gap between them:

pd4ml.generateMulticolumn(2, 8, true);

The appendix starts on a new page because of the *<pd4ml:page.break multicolumn="true"/>*tag in front of it. A page break without the attribute would only move on to the next column.

See H008MulticolumnNewsletter.java in the pd4ml-examples repository for the complete source.